

Elena Vasquez

Senior Backend Engineer | Government Technology

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EXPERIENCE

Senior Systems Engineer | Texas Department of Public Safety | Austin, TX | March 2023 – Present

- Architected autonomous agent pipeline for license plate recognition across 847 field locations, processing 2.3M images daily with 94% first-pass accuracy
- Integrated Claude API with custom state machine for document verification workflows; reduced manual review queue from 180k to 12k monthly items
- Built distributed PostgreSQL + Redis caching layer supporting sub-200ms response times across geographically dispersed nodes
- Implemented retry logic and dead-letter queues for agentic workflows, achieving 99.7% task completion rate
- Optimized inference costs by 41% through prompt engineering and token batching strategies

Software Engineer II | Veterans Affairs Technology Services | Denver, CO | July 2021 – February 2023

- Developed backend services for benefits claims processing system used by 890k monthly active users
- Engineered LLM-powered claim summarization agent that processes 15k documents per day; integrated with legacy SOAP APIs through custom adapter layer
- Managed Kubernetes cluster scaling policies to handle 40% seasonal traffic spikes without exceeding budget thresholds
- Maintained data pipeline ETL jobs (Python, Airflow) transforming raw claims data into training datasets for downstream agents

Backend Engineer | Oregon State Health Authority | Portland, OR | August 2019 – June 2021

- Designed API gateway with request routing logic for 4 internal microservices handling provider registry data
- Built monitoring dashboards and alerting rules; reduced mean time to incident detection from 35 minutes to 8 minutes
- Wrote database migration scripts for PostgreSQL schema updates affecting 120M+ records with zero downtime

Junior Software Engineer | Seattle Public Utilities | Seattle, WA | January 2018 – July 2019

- Developed backend services for meter reading aggregation platform processing 8.7M data points weekly